PRISM-LIVO: Projectively Stable LiDAR-Inertial-Visual Odometry with Localizability-Informed Regularization

Supplementary Material

The supplementary material provides additional details supporting the main text, including:

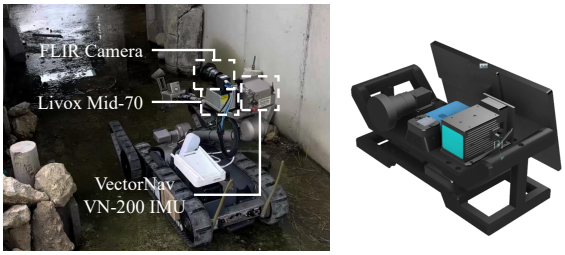
- SA. Platform information for the datasets used in our experiments
- SB. Performance under standard, non-degenerate conditions, showing that our method performs comparably to FAST-LIVO2
- SC. Visualizations of plane-induced homographies under varying view changes
- SD. Justification of the squareness metric, including a comparison against the inverse condition number
- SE. Analysis of photometric residuals at intermediate pyramid levels, beyond the coarsest level
- SF. Motivation and formal derivation of the localizability-informed regularization term
- SG. Runtime results of the proposed system
- SH. Ablation of the visual frontend and of the localizability-informed regularization
- SI. Discussion of limitations in existing LIVO systems and in our proposed approach

SA. Experimental Platforms and Dataset Configurations

We constructed the private dataset [1], [2] using two distinct sensing platforms to evaluate the robustness of the proposed system under diverse operational conditions. The datasets will be publicly released upon acceptance of the associated project paper.

(a) Packbot-based Disaster Inspection Platform. We built a mobile robotic platform based on a PackBot-type ground vehicle for disaster inspection applications. Data were collected in safety-oriented environments designed to emulate confined and partially structured spaces commonly encountered in disaster response scenarios. The overall sensor configuration of this platform is illustrated in Fig. S1(a). The private dataset used in this work consists of two representative sequences from [1]: `IOI-MID70` and `OIO-MID70`. The `IOI-MID70` sequence follows an indoor–outdoor–indoor traversal pattern, whereas `OIO-MID70` follows an outdoor–indoor–outdoor trajectory. These motion patterns were designed to induce substantial illumination variation as well as geometric transitions across different structural environments.

(b) Handheld Platform Configuration. The dataset depicted in Fig. 6 of the main paper was collected using a handheld sensing platform [2]. Its sensor configuration follows the same setup as the Livox Avia-based system depicted in Fig. S1(b). In addition to this sequence, we collected an extra sequence named `handheld_dark` [2]. For this sequence, the camera aperture was intentionally reduced to severely limit incoming light, resulting in extremely low image contrast and severely degraded visual observability. Under such conditions, photometric information is dominated by sparse illumination responses rather than reliable texture cues. Due to the substantial degradation of visual constraints, all compared baselines as well as our proposed method diverged in this sequence. Therefore, `handheld_dark` was excluded from the quantitative evaluation presented in the main paper.



(a) Packbot (b) Handheld

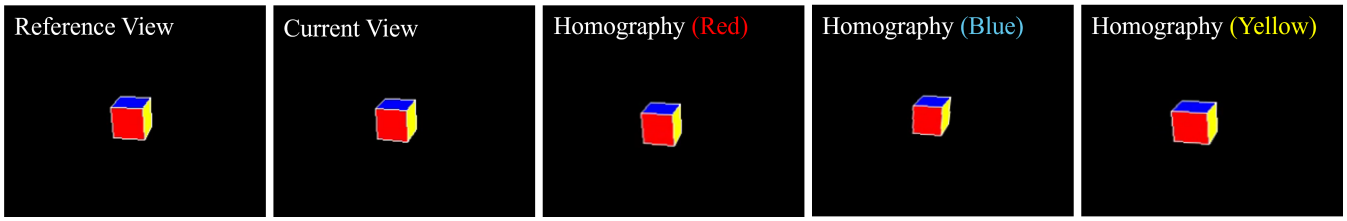
Fig. S1: Sensor configurations of the two platforms used to record the private dataset. (a) PackBot-type ground vehicle, used for the `IOI-MID70` and `OIO-MID70` sequences [1]. (b) Livox Avia-based handheld platform, used for the `handheld_light` and `handheld_dark` sequences [2].

TABLE S1: Absolute trajectory error (ATE) under standard conditions: RMSE ↓ [m] / maximum ↓ [m].

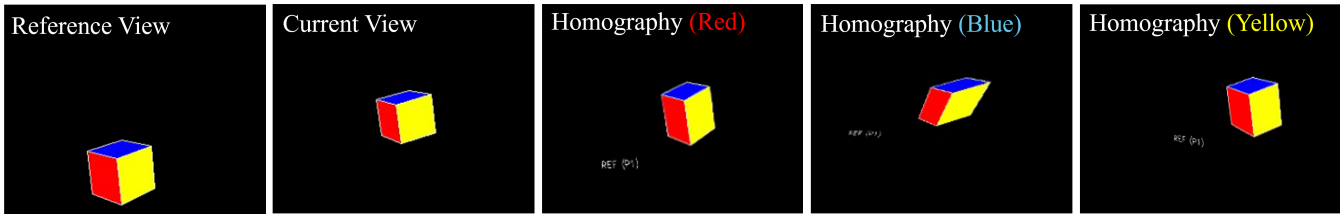
Dataset	**Seq.**	FAST-LIVO2		Ours w/o Reg.		**Ours**	
		RMSE	Max	RMSE	Max	RMSE	Max
NTU-VIRAL [3]	`eee_01`	0.26	0.52	0.27	0.51	0.27	0.51
	`eee_02`	0.21	0.54	0.20	0.54	0.20	0.51
	`eee_03`	0.26	0.46	0.28	0.46	0.29	0.45
	`nya_01`	0.24	0.51	0.24	0.51	0.24	0.51
	`nya_02`	0.23	0.51	0.23	0.51	0.23	0.51
	`nya_03`	0.18	0.51	0.18	0.51	0.18	0.51
	`sbs_01`	0.25	0.40	0.25	0.41	0.25	0.40
	`sbs_02`	0.28	0.50	0.28	0.50	0.28	0.49
	`sbs_03`	0.26	0.54	0.26	0.52	0.26	0.53
Fusion PortableV2 [4]	`escalator00`	0.16	0.56	0.16	0.56	0.16	0.56
	`grass00`	0.09	0.67	0.10	0.74	0.09	0.70
	`room00`	0.09	0.15	0.07	0.15	0.07	0.14

SB. Quantitative Results under Standard Conditions

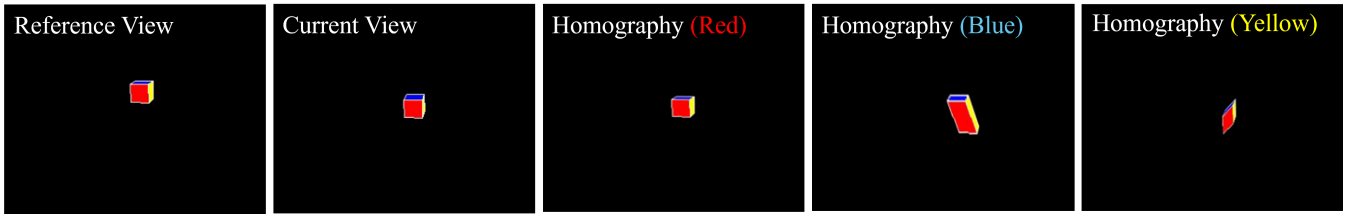
To verify that the proposed modifications do not degrade accuracy in well-conditioned settings, we evaluate on two



(a) Low inter-frame motion



(b) Medium inter-frame motion



(c) High inter-frame motion

Fig. S2: Simulation of plane-induced homographies under increasing inter-frame motion. From top to bottom: low, medium, and high relative pose changes. For each motion level, homographies are independently computed and visualized for the three visible faces of the cube, highlighting plane-dependent geometric distortions.

standard, non-degenerate datasets with reliable ground truth, NTU-VIRAL [3] and FusionPortableV2 [4]. These two are chosen for complementary reasons: NTU-VIRAL has already been evaluated by FAST-LIVO2 on an aerial (drone) platform, enabling a direct comparison on data the baseline itself reports, whereas FusionPortableV2 is selected, among open datasets, to additionally validate our method on a handheld platform. We compare against the original FAST-LIVO2, the baseline our method builds upon, and additionally report our variant without regularization (Ours w/o Reg.) to confirm that the regularization is not detrimental in these settings. As summarized in Table S1, PRISM-LIVO performs on par with FAST-LIVO2 across all sequences, with differences within a few centimeters and no systematic degradation, while the regularized and unregularized variants remain nearly identical. These results confirm that the proposed distortion-aware selection and regularization preserve accuracy when LiDAR geometry is well constrained.

SC. Visualization of Plane-Induced Homographies under Viewpoint Changes

In this section, we visualize simulated plane-induced homographies under varying viewpoint changes using a simple cubic structure. By placing a single cube in the scene and systematically varying the camera viewpoint, we illustrate how increasing inter-frame motion leads to progressively stronger geometric distortions in the induced affine transformations. This controlled simulation highlights the sensitivity of plane-induced homography approximations to large viewangle variations. Although the original method performs local affine transformations derived from plane-induced homographies, the quality of the affine approximation fundamentally depends on the correctness of the underlying homography. If the homography becomes inaccurate due to large viewpoint changes or violations of the planar assumption, the resulting affine transformation inevitably inherits this distortion. Therefore, in this controlled experiment, we visualize the plane-induced homography itself rather than the derived affine transformation to isolate and illustrate the geometric effects of viewpoint variation. To maintain clarity in the comparisons, both the cube-to-camera distance and the magnitude of inter-frame motion were carefully controlled. After multiple simulation trials, we selected three representative scenarios for visualization, corresponding to low, medium, and high inter-frame motion.

As shown in Fig. S2, in the low inter-frame motion case (a), the homographies of all three planes closely resemble the "Current View" reference. In the medium inter-frame motion case (b), the homography exhibits noticeable shearing, while the homography and the homography deviate from the reference but remain within an acceptable range. Finally, in the high inter-frame motion scenario (c), the homography is strongly elongated along one axis, the homography shows stronger shearing, and the homography remains the closest to the reference.

From these observations, three key insights can be drawn.

TABLE S2: Relative motion between the reference and the current view for each panel of Figs. S3–S5. $\|\mathbf{t}\|$ is the baseline and $\Delta\theta$ the relative rotation angle.

Panel	Scene	$\|\mathbf{t}\|$ [m]	$\Delta\theta$ [°]
Fig. S3	Planar wall	1.73	50.17
Fig. S4(b)	1 m voxel, backrest	2.60	65.20
Fig. S4(c)	1 m voxel, seat	1.31	45.53
Fig. S4(d)	0.25 m voxel, same bench	1.08	31.65
Fig. S4(e)	0.25 m voxel, uneven ground	1.78	13.47
Fig. S5(a)	Armrest, first example	1.00	22.34
Fig. S5(b)	Armrest, second example	1.35	58.49
Fig. S5(c)	Box, pose from the estimate	0.10	8.26
Fig. S5(d)	Box, perturbed pose	0.10	8.26
Fig. S5(e)	Tree, viewed frontally	1.14	2.44
Fig. S5(f)	Tree, viewed from the side	2.66	38.04

Panel (d) of Fig. S5 uses the same view pair as (c); the perturbation applied to the pose is 0.375 m in translation and $-2.25°$ in rotation.

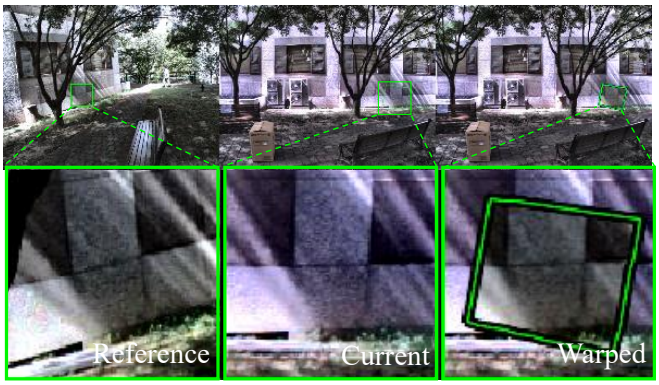


Fig. S3: Plane-induced homography on a genuinely planar surface, computed on real data from a LiDAR-inertial pose estimate. The panels show the reference view, the current view, and the reference patch warped into the current view; the warp remains accurate despite substantial motion between the two views.

(i) Under small inter-frame motion, the effect of selecting the closest view angle for all points within a patch results in a homography that is nearly identical to the real view, producing minimal distortion. (ii) When multiple planes exist within a single patch, the choice of which plane to base the homography on can strongly influence the degree of distortion. This occurs because plane-induced homographies rely on the assumption that all patch points lie on the same plane, an assumption that is easily violated in practice. To mitigate this, we consider multiple candidate points for each patch, aggregating the 26 neighboring voxels along with the central voxel, rather than relying on a single corresponding voxel (see Fig. 2, "Visible Voxel Query" and Sec. III-B). (iii) When examining cases of actual distortion, we observe patterns such as shearing or axis elongation. This observation motivated the design of our squareness metric, which quantifies affine distortion and guides patch selection to minimize geometric deformation. Interestingly, even when many of the transformed homographies exhibit large distortions, the points lying on the same plane are transformed accurately, as highlighted by the homography in Fig. S2(b).

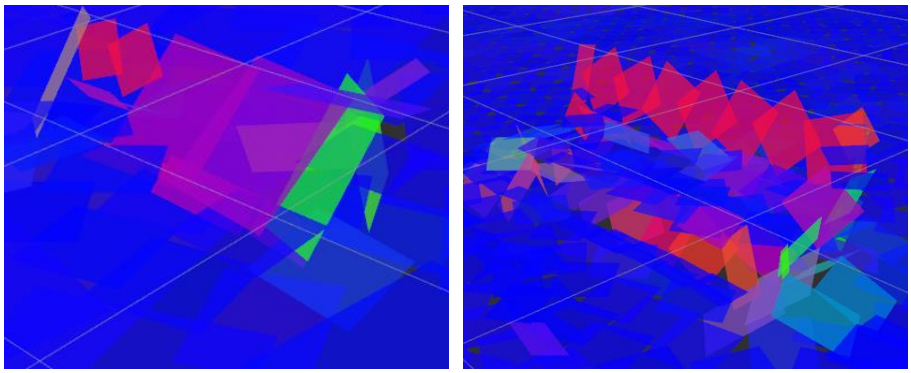
Real data under an accurate pose. The simulation above isolates the effect of viewpoint change with a perfectly known geometry. To see how the homography behaves on real data we acquired a sequence under well-conditioned conditions and computed plane-induced homographies from poses given by a LiDAR-inertial estimate, so that pose error is not the dominant factor. On a genuinely flat surface the plane is easy to estimate and the resulting affine map is accurate, as in Fig. S3, even though the two views are separated by 1.73 m and 50.17° (Table S2). A small positional offset remains, but it is of the order that the photometric optimization itself is able to absorb.

We then examine how sensitive the warp is to four factors in turn: the accuracy of the estimated normal, the choice of plane when several are present, the accuracy of the pose, and the departure of the surface from planarity.

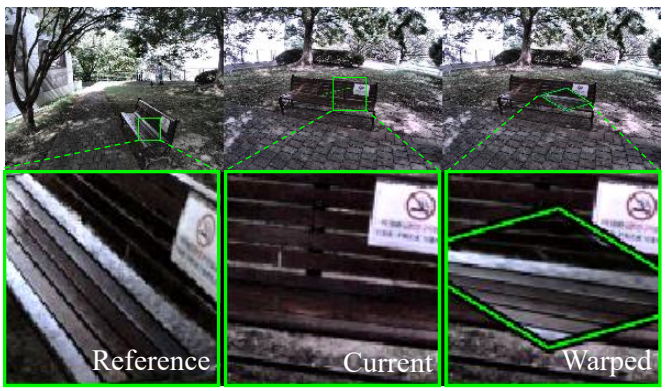
(1) Accuracy of the estimated normal. We vary the voxel size used to build the plane map, at 1.0 m and 0.25 m, and compare the warps obtained on the same bench. The two maps are shown in Fig. S4(a). A bench should ideally be separated into a seat and a backrest, but at 1.0 m the two are fitted as a single plane. At 0.25 m they are separated, although the ground plane is then somewhat uneven and the armrest is absorbed into a neighboring voxel; both matter below. With the coarse map, Fig. S4(b) and (c), the warp is still usable in both cases despite the inaccurate normal, and (b) is visibly the cleaner of the two. This is consistent with the map: the single fitted normal lies much closer to the normal of the backrest, which is the surface (b) observes, than to that of the seat. With the finer map the normal on the bench is estimated far better, and the affine map of Fig. S4(d) is accurate. The residual effect of the uneven ground is visible in Fig. S4(e), where the warp is noticeably distorted.

(2) Choice of plane when several are present. Fig. S5(a) and (b) warp the same bench and are severely distorted. The distortion is not explained by the viewpoint: (a) is separated by only 1.00 m and 22.34°, less than any of the panels of Fig. S4. What differs is the plane. The region actually visible in the patch is mostly backrest and seat, while the normal used comes from the armrest, so the homography is built on a surface the patch does not lie on. This reproduces, on real data, the effect of the controlled simulation above.

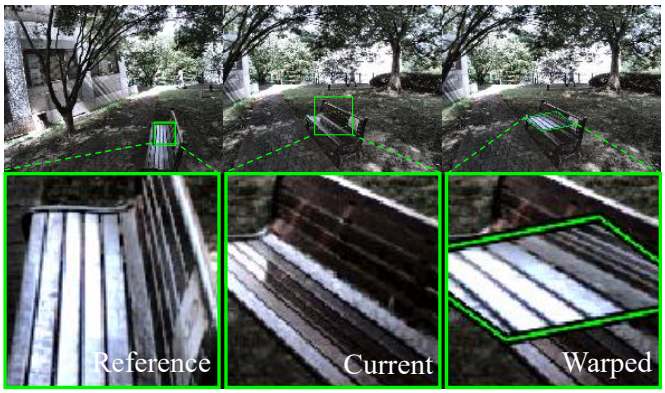
(3) Accuracy of the pose. Odometry error accumulates over time, so we also ask how much of the warp quality depends on the state estimate rather than on the plane. Fig. S5(c) is taken from a nearly identical viewpoint, 0.10 m and 8.26°, and the affine map is close to exact. Applying the perturbation of Table S2 to the translation and the rotation produces the badly misaligned warp of Fig. S5(d); the injected translation, 0.375 m, is larger than the baseline between the two views. This is not a clean experiment, since a displaced pose would also change which voxels are queried in the first place, but it shows that pose error alone is enough



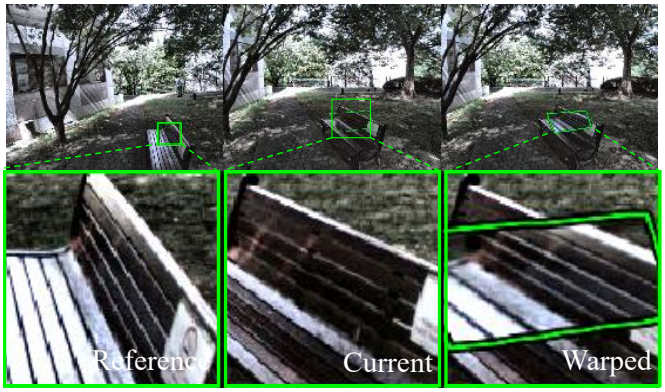
(a) Voxel map of the same bench at 1 m (left) and 0.25 m (right)



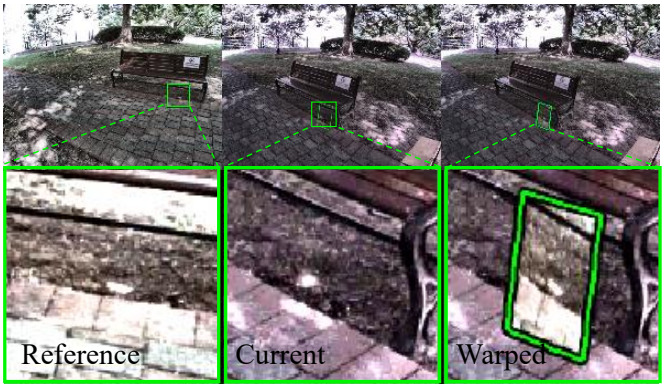
(b) 1 m voxel, backrest



(c) 1 m voxel, seat



(d) 0.25 m voxel, same bench



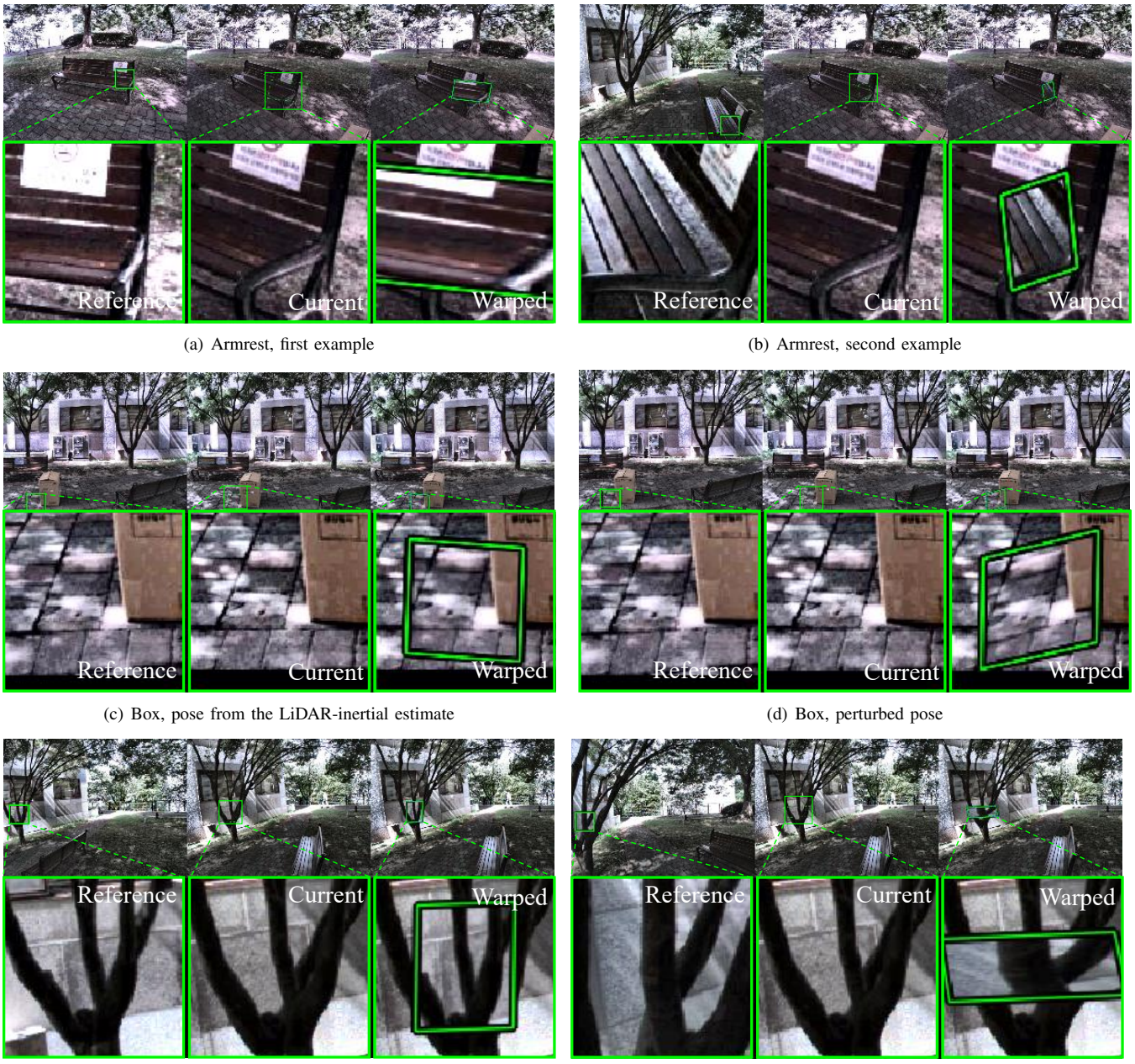
(e) 0.25 m voxel, uneven ground

Fig. S4: Effect of the voxel resolution on the plane-induced homography, on the same bench and the same pair of views. A bench is two rectangles meeting at an angle; at 1 m the two are fitted as a single plane, at 0.25 m they are separated. (a) shows the two voxel maps. Panels (b)–(e) each show the reference view, the current view, and the reference patch warped into the current view: (b) and (c) use the coarse map and warp the backrest and the seat, respectively, (d) uses the fine map on the same bench, and (e) uses the fine map on the uneven ground.

to corrupt the warp substantially.

(4) Departure from planarity. Finally we consider a surface that is not planar at any resolution. A tree branch is planar enough for the LiDAR-inertial stage to use it for pose estimation, but its surface is close to a cylinder, so the visually correct plane normal changes rapidly with the viewpoint. Under a small viewpoint change, 1.14 m and 2.44°, the warp of Fig. S5(e) is not badly distorted. Under a larger one, 2.66 m and 38.04°, the warp of Fig. S5(f) is dominated by the error in the plane normal. With only one surface in play here, keeping the warp undistorted reduces to keeping the viewpoint change small, the same quantity a closest-view criterion ranks on directly.

What these experiments imply. Taken together, they show that shear or elongation in the affine matrix is not by itself evidence that the resulting measurement is wrong: several of the warps above are strongly sheared and still agree with the current view. They also show the converse, which is the point we wish to make. The presence of several planes within a patch, an error in the estimated pose, and a surface that



(a) Armrest, first example

(b) Armrest, second example

(c) Box, pose from the LiDAR-inertial estimate

(d) Box, perturbed pose

(e) Tree, viewed frontally

(f) Tree, viewed from the side

Fig. S5: Failure modes of the plane-induced homography that remain once the voxel resolution is fine enough: a surface perpendicular to the fitted plane (a, b), an inaccurate pose (c, d), and a genuinely non-planar structure (e, f).

is not planar can each produce a warp that is geometrically plausible and yet does not describe what the camera sees, and a photometric residual computed from such a patch will mislead the state update. None of these conditions is visible in the affine matrix, and none can be detected by the criterion we propose. What can be done is to limit the exposure to them. A patch whose warp is close to the identity is the one that relies least on the plane it was derived from, so when the underlying model may be wrong, selecting the least distorted patch — which is what the squareness score does — is the conservative choice. This is the sense in which we regard projective stability as a safeguard rather than as a correction.

These observations highlight the geometric effects of viewpoint variation and motivate our use of squareness-aware patch selection, which aims to preferentially choose patches that are less susceptible to affine distortions.

We also note that our approach still rests on this local planar assumption: the plane-induced homography, and hence the squareness computed from it, is valid only when a patch corresponds to a locally planar surface. While the multi-candidate selection above mitigates the common case of multiple planes within a patch, genuinely non-planar or finely structured regions violate the assumption, and selecting the least-distorted affine cannot fully compensate for the

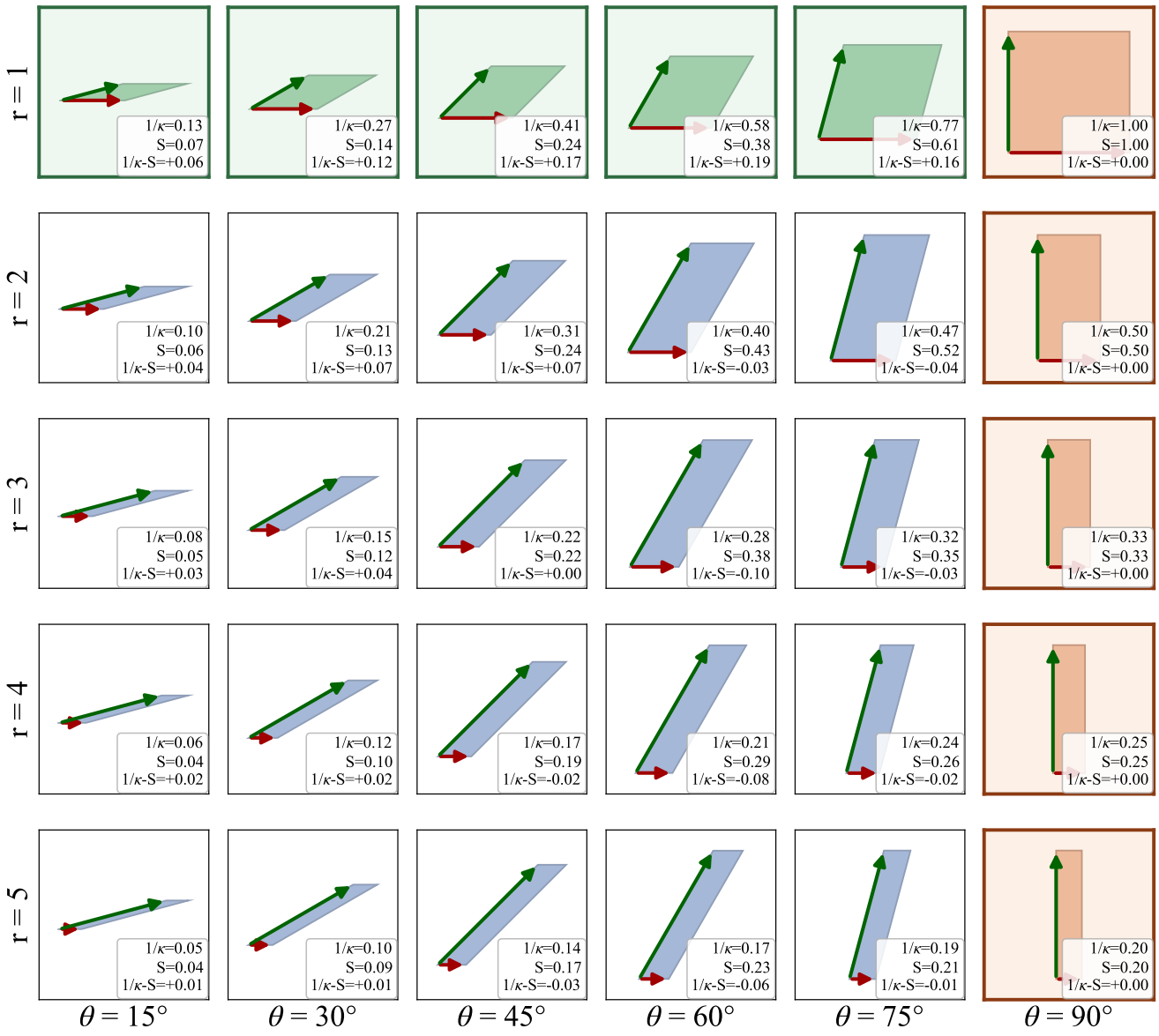


Fig. S6: Comparison between the proposed squareness score S and the inverse condition number $1/\kappa$ for affine warps, parameterized by the basis-vector length ratio $r = \|\mathbf{a}_2\|/\|\mathbf{a}_1\|$ and the angle θ between them.

resulting model error.

SD. Justification of the Squareness Metric

None of the representative sparse direct methods evaluates the distortion of the warp that its selection produces: PTAM [5] performs no selection among views at all and matches its single reference patch by the zero-mean sum of squared differences (SSD), DSO [6] does not warp patches explicitly, and SVO [7], FAST-LIVO1 [8], and FAST-LIVO2 [9] rank candidates by viewing geometry or photometric similarity rather than by the resulting affine matrix.

As shown in the controlled simulation of Sec. SC, viewpoint change induces affine distortion in the form of shearing, elongation, or both. Suppressing shearing requires the angle θ between the two basis vectors to stay close to $90°$, while limiting elongation requires their length ratio to stay close to one. Evaluating these two conditions separately is problematic: when one patch shows little elongation but strong shearing and another shows the opposite, two independent measures provide no principled way to rank them. The squareness metric instead folds both effects into a single scalar that directly measures how close the affine warping is to a square ($S = 1$ at no distortion), thereby resolving this ambiguity.

A natural alternative that also measures deviation from a similarity transform is the condition number $\kappa = \sigma_{\max}/\sigma_{\min}$ of the affine matrix, and the two are in fact closely related. For shear-free (rectangular, $\theta = 90°$) warps they coincide exactly ($S = 1/\kappa$), and as Fig. S6 shows, their values remain close in general. They differ most when the basis lengths are

TABLE S3: Reference selection by the closest view angle, by the inverse condition number, and by the proposed squareness score. Absolute trajectory error (ATE): RMSE ↓ [m] / maximum ↓ [m]. The two right-hand columns differ only in the ranking function; the closest-view column additionally lacks the enlarged candidate search and the robust weighting. The best value of each row is shown in bold.

Sequence	Closest-view w/ reg.		Inv. cond. number $1/\kappa$		**Squareness** S	
	RMSE	Max	RMSE	Max	RMSE	Max
`Bridge01`	**67.47**	**163.69**	69.84	172.22	69.84	172.44
`UrbanTunnel02`	291.19	412.67	107.26	197.49	**99.57**	**179.35**
`UrbanTunnel03`	298.72	562.81	265.99	**493.29**	**260.41**	497.95
`Cave01`	×	×	0.42	2.73	**0.30**	**2.19**
`Cave02`	**0.22**	**0.60**	0.27	0.76	0.25	0.72
`IOI-MID70`	0.25	**1.01**	**0.20**	1.24	**0.20**	1.20
`OIO-MID70`	0.75	1.91	**0.25**	**1.40**	**0.25**	1.45

balanced ($r = 1$) and distortion is dominated by shear: there the gap $1/\kappa - S$ is positive, i.e., S responds slightly more strongly to shear, though the difference again vanishes as the shear grows large. Because the two measures are this similar, we do not claim that squareness is a superior distortion measure; our claim is rather the need for projectively stable patch selection itself, with squareness serving as one simple and effective way to realize it. A condition-number-based selection would be expected to perform comparably. Table S3 verifies this empirically: with every other component held fixed, we replace the squareness score by the inverse condition number as the ranking function. The two criteria are indeed close: on `Bridge01`, `IOI-MID70` and `OIO-MID70` they reach the same RMSE to two decimal places, confirming that a condition-number-based selection is a viable alternative. On the remaining four sequences the inverse condition number is nevertheless slightly worse, and we see two plausible reasons.

The first is the rejection threshold. We reuse the value $\tau_{sq} = 0.5$ that was chosen for the squareness score, which need not be appropriate for a different measure. Table S5 supports this reading: on three of the four sequences the RMSE obtained with the inverse condition number (107.26 m, 265.99 m and 0.27 m on `UrbanTunnel02`, `UrbanTunnel03` and `Cave02`) falls inside the range spanned by sweeping τ_{sq} for the squareness score, so a gap of this size is within what the threshold alone can produce; only `Cave01` (0.42 m) lies above that range.

The second is the different way the two measures respond to shear and to elongation, visible in Fig. S6. At $r = 1$, where the warp is a pure shear, S lies below $1/\kappa$ for every θ, so the squareness score penalizes shear more strongly. Conversely, for $r \geq 2$ under moderate shear ($\theta = 60°$ and $75°$) S exceeds $1/\kappa$, so it is the more permissive of the two towards elongation. This ordering may be the favorable one for patch selection, in that a shear tilts the patch as a whole whereas an elongation distorts it along a single axis only, which would make the former the more damaging of the two for photometric alignment; we offer this as an interpretation rather than as something the experiments establish.

Tail behavior of the photometric residuals. The trajectory error of Table S3 aggregates over an entire sequence and is therefore insensitive to how often a badly distorted patch is selected. Since the criterion is intended to suppress severe distortions rather than to lower the typical residual, we also measure the tail of the residual distribution directly. For each frame we take the ratio of the 95th percentile of the patch residuals to their median, and report the median of that ratio over the sequence together with the fraction of frames in which it exceeds four. Both quantities are scale-free, so they can be compared across sequences of very different error magnitude. Table S4 reports them for the four criteria.

The separation is much clearer here than in the trajectory error. On `UrbanTunnel02` the ratio is 83.64 for *closest-view* selection against 6.56 for the *squareness* score, a factor of 12.7, and on `Bridge01` it is 32.05 against 8.19. The *NCC* criterion of FAST-LIVO2 behaves similarly, at 23.13 and 13.70 on `Bridge01` and `UrbanTunnel03`. The fraction of frames with a heavy tail follows the same ordering, dropping from 92–99% for the two baseline criteria, *NCC* and *closest-view*, to 62–82% for the two distortion-aware ones, *squareness* and the *inverse condition number*.

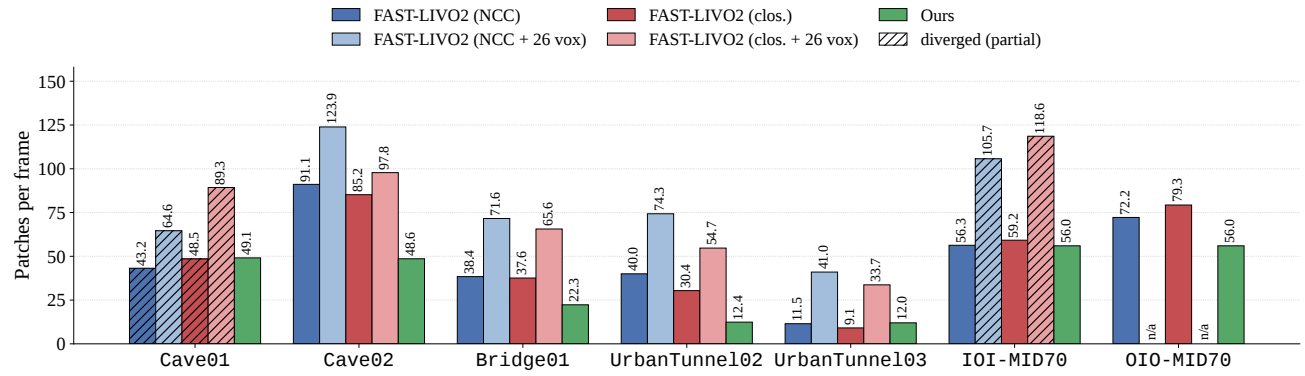
The two distortion-aware criteria are again close to one another, as the analysis above predicts: *squareness* is the lower of the two on `UrbanTunnel02` and `UrbanTunnel03`, the *inverse condition number* on `Bridge01`, `Cave01` and `Cave02`, and they are identical on `IOI-MID70`. What separates them from the baselines is not the typical residual but how often an extreme one occurs, which is the quantity the criterion was designed to control. `OIO-MID70` is the one sequence in which this ordering does not hold: *closest-view* selection attains the lowest ratio there, at 15.42 against 22.72 and 21.77. The fraction of frames with a heavy tail is nevertheless the same for all three, within 0.2 percentage points of 88%, so the three criteria are better described as comparable on this sequence than as separated by it.

The squareness score is also cheaper to evaluate than the condition number, since it requires no singular value decomposition. In our C++ implementation, evaluating 2,000 candidate matrices takes 98.709 μs with an SVD (as required by the condition number) versus 20.996 μs with squareness, i.e., squareness is about 4.7× faster. In absolute terms, however, the difference is very small—both are well within 0.1 ms—and it was measured on a high-end desktop (see Sec. SG), so we do not put it forward as an argument in favor of the metric. It may nonetheless matter on an embedded platform, which we leave to the user.

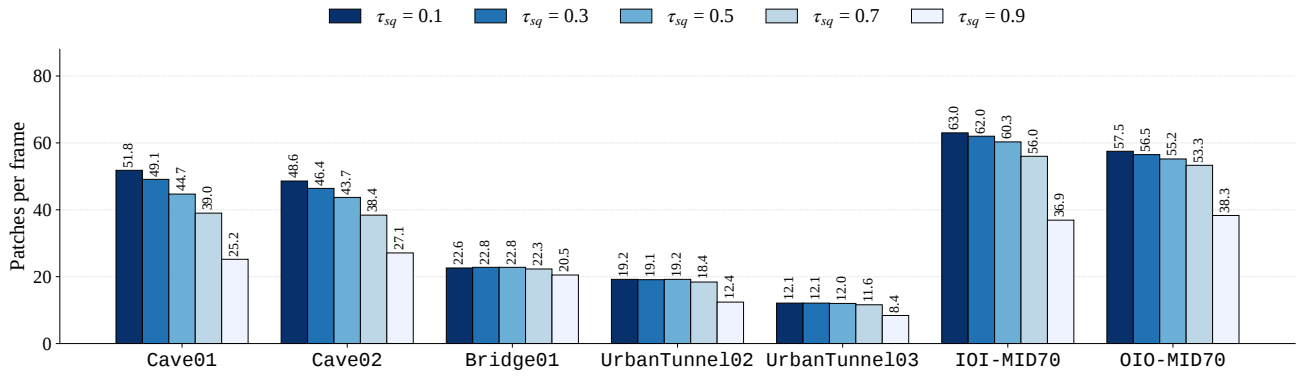
Finally, we clarify how the thresholds of the four outlier-rejection criteria in Sec. III-C of the main paper are chosen. The view-angle and normal-angle thresholds are taken directly from prior work [7]–[9]. The squareness threshold is set heuristically: from the visual examples in Fig. S6, $S \approx 0.6$ marks the level below which the affine warp appears clearly distorted, and we relax this slightly to a more permissive $S < 0.5$ so as not to discard too many measurements; most selected patches already have S close

TABLE S4: Tail behavior of the photometric residuals under four reference-selection criteria. For each frame we form the ratio of the 95th percentile of the patch residuals to their median; $\mathrm{p95/med}$ is the median of that ratio over the sequence, and $\mathrm{p95} > 4\,\mathrm{med}$ is the fraction of frames in which the ratio exceeds four. Lower is better in both columns, and the best value of each row is shown in bold. $\times$ denotes a diverged run.

Sequence	FAST-LIVO2 (NCC)		Closest-view		**Squareness**		Inv. cond. number	
	p95/med	p95 > 4 med [%]	p95/med	p95 > 4 med [%]	p95/med	p95 > 4 med [%]	p95/med	p95 > 4 med [%]
Bridge01	23.13	98.3	32.05	99.2	8.19	74.9	**7.86**	**74.1**
UrbanTunnel02	35.22	93.9	83.64	97.4	**6.56**	**65.6**	7.72	77.6
UrbanTunnel03	13.70	93.2	20.74	92.1	**5.17**	**62.1**	5.37	63.6
Cave01	×	×	×	×	6.64	78.8	**5.86**	**70.9**
Cave02	9.95	92.2	11.29	90.0	8.28	81.9	**6.61**	**71.7**
IOI-MID70	–	–	12.38	84.7	**9.12**	77.7	**9.12**	**77.6**
OIO-MID70	–	–	**15.42**	88.2	22.72	88.1	21.77	**88.0**



(a) Patches retained by each configuration



(b) Patches retained as the squareness threshold τ_{sq} is swept

Fig. S7: Average number of patches retained per frame. Hatched bars in (a) denote runs that diverged, for which the count is taken over the segment before divergence.

to one, so this removes only a small fraction of severely distorted patches (Fig. S7(b)). Since this criterion overlaps in effect with the view- and normal-angle filters, we expect its exact value to have limited impact, and we accordingly keep $S < 0.5$ fixed across all datasets and platforms. That said, setting the threshold close to one rejects many measurements, and the most suitable value may vary across datasets; we therefore suggest that users adjust it for their own setting. Table S5 reports a sweep of τ_{sq} from 0.1 to 0.9 with all other components fixed.

SE. Photometric Residual Analysis beyond the Pyramid Level

While the main text reports photometric residuals only at the coarsest level, here we provide a statistical analysis across all pyramid levels to offer a more comprehensive evaluation (see Sec. IV-C and Fig. 7 for the metrics and results at the coarsest level).

As shown in Table S6, as the image pyramid level increases, the photometric residuals produced by our method tend to be lower, and this reduction is statistically significant. However, at level 0, there are cases where the baselines

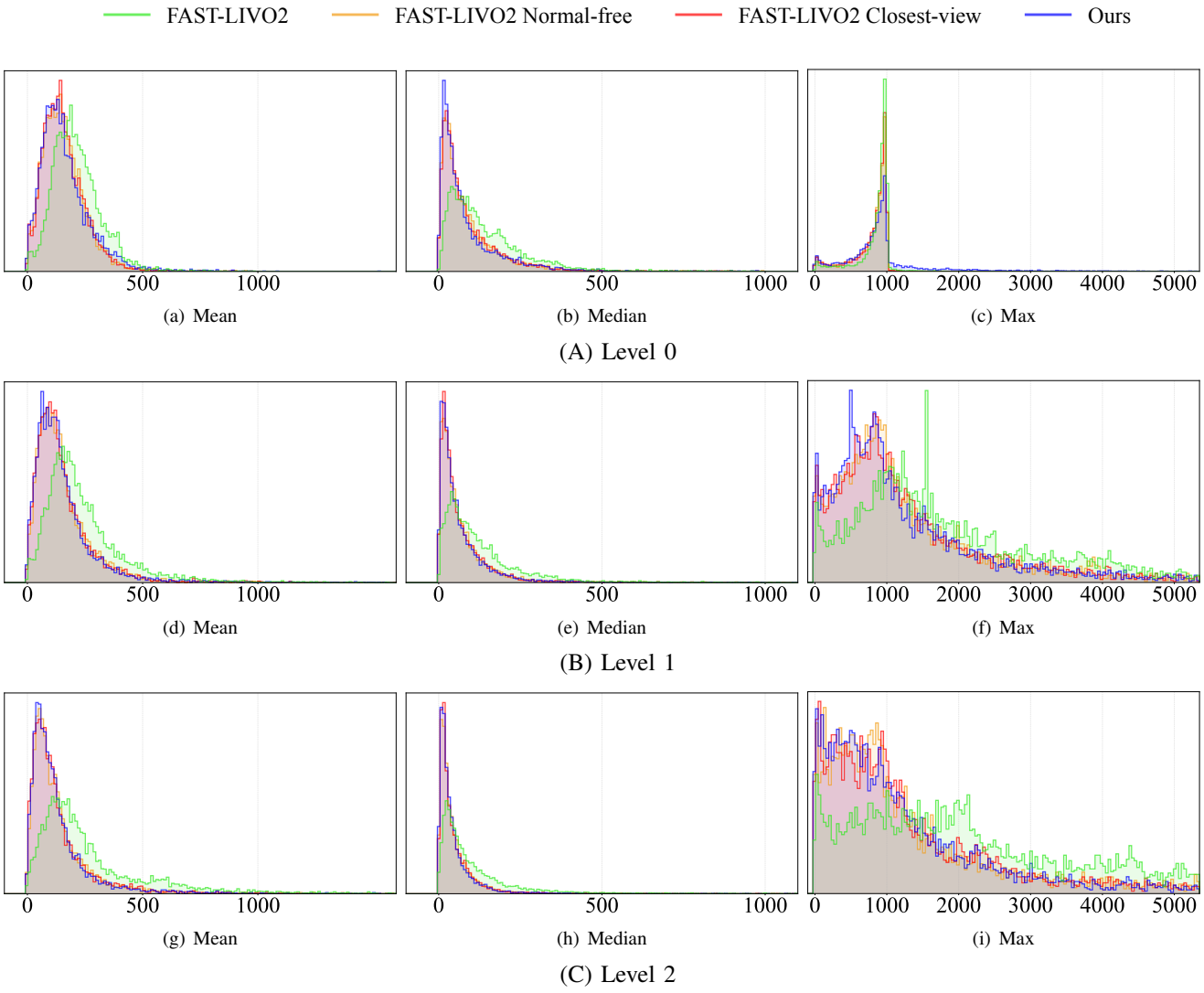


Fig. S8: Overall statistical distribution of photometric errors across different levels. The x-axis represents the magnitude of patch-wise photometric errors, and the y-axis denotes the probability density.

TABLE S5: Sensitivity of the squareness rejection threshold τ_{sq}: a patch is discarded when $S < \tau_{sq}$, so a larger τ_{sq} rejects more measurements. Absolute trajectory error (ATE) RMSE $\downarrow$ [m]. All other components are held fixed, and $\tau_{sq} = 0.5$ (underlined) is the value used throughout the paper. The best value of each row is shown in bold.

Sequence	τ_{sq}				
	0.1	0.3	<u>0.5</u>	0.7	0.9
`Bridge01`	**69.57**	70.31	69.84	69.68	69.70
`UrbanTunnel02`	104.87	107.87	99.57	95.88	**87.15**
`UrbanTunnel03`	271.34	263.22	**260.41**	265.89	272.11
`Cave01`	0.31	0.33	**0.30**	0.39	0.39
`Cave02`	0.29	0.28	0.25	**0.24**	0.26
`IOI-MID70`	0.22	0.21	0.20	**0.18**	0.20
`OIO-MID70`	0.30	**0.22**	0.25	0.24	0.23

exhibit lower photometric residuals. Regarding the mean and median, the differences are very small, indicating that the actual discrepancy is not substantial. For the maximum, however, the absolute value of the t-statistic appears large. As shown in Fig. S8(A-c), this is because our method exhibits longer tails, which are truncated at level 0 due to filtering of photometric residuals exceeding 1,000. These tails extend beyond 1,000 because filtering and residual computation use different tools: filtering is performed with ViKit, whereas the residuals are computed via the Jacobian, which directly calculates the interpolation coefficients. Interestingly, for all metrics, the distribution of the original FAST-LIVO2 [9] shifts toward larger values as the pyramid level increases. This can be attributed to the fact that the original FAST-LIVO2 does not account for view-change-induced distortions, which become more pronounced as patch size increases.

SF. Motivation and Formal Derivation of the Localizability-Informed Regularization

This regularization strategy is motivated by [10], a LiDAR-based odometry method that suppresses state updates along poorly observable (low-localizability) directions during LiDAR state estimation. In contrast, our approach leverages

TABLE S6: Statistical significance analysis: t-test results (t / p) for various scaling functions.

Level	Mean (t / p)			Median (t / p)			Max (t / p)		
	FAST-LIVO2 Normal-free	FAST-LIVO2	FAST-LIVO2 Closest-view	FAST-LIVO2 Normal-free	FAST-LIVO2	FAST-LIVO2 Closest-view	FAST-LIVO2 Normal-free	FAST-LIVO2	FAST-LIVO2 Closest-view
0	**-3.35 / 0.00**	**23.63 / 0.00**	**-4.49 / 0.00**	1.74 / 8.26	**26.15 / 0.00**	1.03 / 30.50	**-17.16 / 0.00**	**-14.66 / 0.00**	**-16.98 / 0.00**
1	1.54 / 12.44	**25.59 / 0.00**	0.09 / 92.44	1.74 / 8.15	**18.56 / 0.00**	1.05 / 29.36	-0.92 / 35.25	**13.69 / 0.00**	**-3.34 / 0.04**
2	**2.58 / 0.01**	**28.34 / 0.00**	1.12 / 26.33	**1.99 / 0.04**	**13.22 / 0.00**	1.32 / 18.77	0.59 / 55.32	**33.40 / 0.00**	1.23 / 22.04

* **Blue** indicates a statistically significant **lower** photometric residual ($p < 5\%, t > 0$) compared to the baseline.
* **Red** indicates a statistically significant **higher** photometric residual ($p < 5\%, t < 0$) compared to the baseline.

LiDAR localizability during the visual update step, enforcing regularization along directions of high localizability where LiDAR provides strong geometric constraints. As a result, the visual update is guided to remain consistent with the well-constrained LiDAR geometry. We now derive the formulation in detail. First, at the LIS stage, the Jacobian of the error with respect to the final iteration of the IESKF can be computed as $\mathbf{J}_{\text{LIS}}$. From this, the Hessian matrix can be formed as:

$$\mathbf{H}_{\text{LIS}} = \mathbf{J}_{\text{LIS}}^{\top}\mathbf{J}_{\text{LIS}} = \begin{bmatrix} \mathbf{H}_{\text{rr}} & \mathbf{0}_{3\times3} & \cdots \\ \mathbf{0}_{3\times3} & \mathbf{H}_{\text{tt}} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \in \mathbb{R}^{20\times20}. \quad (1)$$

By performing eigenvalue decomposition on the Hessian blocks $\mathbf{H}_{\text{rr}}$ and $\mathbf{H}_{\text{tt}}$, the corresponding eigenvectors, $\mathbf{V}_{\text{rr}}$ and $\mathbf{V}_{\text{tt}}$, are identified as localizability vectors. Rather than using the eigenvalues directly, we adopt the strong contribution degrees from X-ICP [11] as the localizability degrees, $\mathbf{Q}_{\text{rr}}$ and $\mathbf{Q}_{\text{tt}}$. For notational simplicity in this derivation, $\mathbf{C}$ and $\boldsymbol{\mathcal{Q}}$ are written in their zero-padded form matching the dimension of $\mathbf{x}$, which is equivalent to the compact formulation in the main text since the padded entries contribute nothing to the regularization. Using these, we construct Eq. (9) in the main text, and Eq. (10) can be reformulated as follows:

$$\begin{aligned} \min_{\delta\mathbf{x}} \ & \cdots + \|\mathbf{x} - \widehat{\mathbf{x}}_{\text{LIS}}\|^2_{\widehat{\mathbf{P}}^{-1}_{\text{LIS}}} + \left|\boldsymbol{\mathcal{Q}}^{\top}\mathbf{C}(\mathbf{x} - \widehat{\mathbf{x}}_{\text{LIS}})\right|^2 \\ & = \cdots + (\mathbf{x} - \widehat{\mathbf{x}}_{\text{LIS}})^{\top}(\underbrace{\widehat{\mathbf{P}}^{-1}_{\text{LIS}} + \mathbf{C}^{\top}\boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{Q}}^{\top}\mathbf{C}}_{\mathbf{P}_{\text{L}}^{-1}})(\mathbf{x} - \widehat{\mathbf{x}}_{\text{LIS}}). \end{aligned} \quad (2)$$

Next, by incorporating the Jacobian of the true state with respect to the error state at the ℓ-th iteration, $\mathbf{J}^{\ell}_{\delta}$, the covariance is updated as:

$$(\mathbf{P}^{\ell})^{-1} = (\mathbf{J}^{\ell}_{\delta})^{\top}\mathbf{P}_{\text{L}}^{-1}(\mathbf{J}^{\ell}_{\delta}). \quad (3)$$

Here, $\mathbf{J}^{\ell}_{\delta}$ and the remaining IESKF update equations are provided in [12] (note that the notations used here slightly differ from those in [12]; for instance, the Jacobian is denoted as $\mathbf{H}$ in [12]).

For a simple interpretation, let us assume that $\boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{Q}}^{\top}$ is a diagonal matrix:

$$\boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{Q}}^{\top} = \boldsymbol{\mathcal{Q}}_{\text{D}} = \text{diag}(\lambda_{r_1}, \lambda_{r_2}, \ldots, \lambda_{t_3}, 0, \ldots, 0). \quad (4)$$

Then, we have

$$\mathbf{C}^{\top}\boldsymbol{\mathcal{Q}}_{\text{D}}\mathbf{C} = \lambda_{r_1}\mathbf{v}_{r_1}\mathbf{v}_{r_1}^{\top} + \lambda_{r_2}\mathbf{v}_{r_2}\mathbf{v}_{r_2}^{\top} + \ldots + \lambda_{t_3}\mathbf{v}_{t_3}\mathbf{v}_{t_3}^{\top}, \quad (5)$$

TABLE S7: Runtime Comparison [ms] ↓

Sequence	FAST-LIVO2	Ours
Bridge01	9.15	13.92 (↑ 4.77)
UrbanTunnel02	–	15.34 (–)
UrbanTunnel03	11.79	17.18 (↑ 5.39)
Cave01	–	11.35 (–)
Cave02	7.07	9.56 (↑ 2.49)
IOI-MID70	6.41	8.55 (↑ 2.14)
OIO-MID70	7.69	10.41 (↑ 2.72)

which corresponds to projecting $\mathbf{x} - \widehat{\mathbf{x}}_{\text{LIS}}$ onto the subspace spanned by each localizability vector and scaling according to the localizability degree. In other words, our method increases the magnitude in the high-localizability subspace to enforce stronger regularization, whereas in the low-localizability subspace, the regularization values are nearly zero, allowing the default visual optimization to proceed.

Of course, assuming that the LiDAR can provide sufficient constraints in all directions, a switching system, as in [13], can be used to trigger updates only in cases of LiDAR degeneracy. However, this approach can be susceptible to false positives. Therefore, while such a switching system may be more robust in most scenarios, our contribution lies in exploring an alternative approach through a new method.

SG. Runtime and Computational Complexity Analysis

We measured the average runtime across all datasets used in our experiments, except for cases where FAST-LIVO2 diverged. Experiments on open datasets were conducted on a desktop equipped with an Intel Core i9-13900KF CPU (24 cores, 32 threads, 5.8 GHz), an NVIDIA GeForce RTX 3090 Ti GPU, 32 GB RAM, and Ubuntu 20.04 LTS. As shown in Table S7, our method exhibits a higher runtime. This is because, unlike the original FAST-LIVO2, which updates a reference patch only when its pixel location changes significantly, our method searches for one in every frame, resulting in the observed increase in runtime. Furthermore, our method projects all points contained in the 27 voxels corresponding to a single grid cell, rather than just one point, which also contributes to this difference. However, the differences in runtime remain within 5.4 ms, and the total runtime itself is still well within the 100 ms LiDAR cycle, making it sufficiently fast for practical use. It should be noted, however, that our tests were conducted on a high-end desktop, and additional validation would be necessary for embedded platforms.

TABLE S8: Effect of the enlarged candidate search on the two baseline selection criteria. Absolute trajectory error (ATE): RMSE ↓ [m] / maximum ↓ [m]. The proposed squareness criterion is evaluated in Table III of the main paper, on the same candidate set as the C row with the enlarged search enabled. The top three results in each column are highlighted as **first**, second, and third.

Configuration		GEODE [14]						Superloc [15]				Private			
Patch sel.	27 Voxel	Bridge01		UrbanTunnel02		UrbanTunnel03		Cave01		Cave02		IOI-MID70		OIO-MID70	
		RMSE	Max	RMSE	Max	RMSE	Max	RMSE	Max	RMSE	Max	RMSE	Max	RMSE	Max
N		85.05	205.39	287.33	401.99	297.39	558.99	×	×	**0.22**	**0.64**	18.34	34.18	18.11	33.97
N	✓	75.59	190.68	**276.23**	**393.01**	294.89	562.89	×	×	0.26	0.75	×	×	×	×
C		**68.41**	**166.30**	281.63	397.90	291.29	552.72	×	×	0.26	0.67	**0.28**	**1.06**	**0.94**	**2.57**
C	✓	71.13	172.91	285.76	403.88	**289.59**	**548.26**	×	×	0.28	0.79	8.11	112.45	×	×

N: normal–viewing alignment with normalized cross-correlation (FAST-LIVO2). C: closest view angle.

SH. Ablation of the Visual Frontend and the Regularization

Table III of the main paper ablates the two components that act on the visual update, the robust weighting and the localizability-informed regularization, on top of the proposed frontend. This section ablates the frontend itself. Table S8 isolates the effect of the enlarged candidate search on the two baseline selection criteria, and Table S9 pairs every criterion with and without the regularization.

Enlarged candidate search. Collecting map points from the 26 neighboring voxels does not by itself improve accuracy. It helps the normal–viewing criterion on the three long GEODE sequences, where the RMSE on Bridge01 drops from 85.05 m to 75.59 m, but it degrades or destabilizes every remaining case. Both private sequences diverge, and under closest-view selection IOI-MID70 degrades from 0.28 m to 8.11 m while OIO-MID70 diverges outright. The reason is that a larger candidate set also contains more strongly distorted patches. The number of retained patches roughly doubles on every sequence (Fig. S7(a)), and the additional ones are exactly those that the baseline criteria have no means of rejecting, since neither ranks candidates by the distortion of the warp they induce. Enlarging the candidate set is therefore useful only if the additional candidates are filtered, and not in itself.

Squareness criterion. Comparing the last row of Table S8 with the first row of Table III of the main paper isolates the proposed criterion, since both operate on the same enlarged candidate set and differ only in how a reference view is ranked. Replacing the closest-view ranking by the squareness score improves all seven sequences: from 8.11 m to 0.22 m on IOI-MID70 and from 285.76 m to 113.85 m on UrbanTunnel02, while the two diverging sequences, Cave01 and OIO-MID70, are recovered. The proposed frontend in fact retains the fewest patches of all configurations on most sequences (Fig. S7(a)), so the gain comes from discarding the distorted candidates that the enlarged search introduces, rather than from using more of them.

Localizability-informed regularization. The regularization improves the majority of the reliable entries, and it does so on top of every patch-selection criterion rather than only on ours: on Bridge01 it lowers the RMSE of FAST-LIVO2 from 85.05 m to 75.08 m and that of the closest-view variant from 68.41 m to 67.47 m. Its effect is nevertheless most consistent on the proposed frontend. Without the robust weighting it improves all seven sequences, and with it five of the seven improve while the two exceptions, Cave01 and OIO-MID70, degrade by 0.01 m. This is the behavior the formulation is designed for: the term constrains the visual update toward the LiDAR-inertial prior along the well-localizable directions, which helps only to the extent that the remaining visual correction is itself trustworthy.

These gains are specific to the difficult sequences considered here. In such scenes the LiDAR-inertial stage still constrains a subset of directions well while the visual update is at its least reliable, so transferring that geometric information is worthwhile. Under well-conditioned conditions the term is instead essentially inert: in Table S1 the variants with and without the regularization are identical on ten of the twelve sequences and differ by 0.01 m on the remaining two, giving the same mean ATE RMSE of 0.210 m. The regularization therefore acts where the geometry is informative and the visual update needs constraining, and stays out of the way otherwise.

SI. Limitations and Future Work

Our method addresses one of the main issues in the original FAST-LIVO2, suggesting that distortions in affine transformations can lead to incorrect state updates, and thereby improves robustness. However, it still has five main limitations:

- Reliance on the local planar assumption: the plane-induced homography, and hence the squareness score computed from it, is valid only when a patch corresponds to a locally planar surface. The multi-candidate selection of Sec. III-C of the main paper mitigates the common case of several planes within a patch, but genuinely non-planar or finely structured regions violate the assumption, and selecting the least distorted affine matrix cannot compensate for the resulting model error. As Sec. SC shows, an inaccurate pose or a non-planar surface can each produce a warp that is geometrically plausible and yet does not describe what the camera sees, and no criterion acting on the affine matrix alone can detect this.

TABLE S9: Ablation of the localizability-informed regularization. Absolute trajectory error (ATE): RMSE ↓ [m] / maximum ↓ [m]. Each pair of rows differs only in whether the regularization is applied, and the change relative to the preceding row is given as improvement ↓ / degradation ↑. The best value of each column is shown in bold.

Method	GEODE [14]						Superloc [15]				Private			
	`Bridge01`		`UrbanTunnel02`		`UrbanTunnel03`		`Cave01`		`Cave02`		`IOI-MID70`		`OIO-MID70`	
	RMSE	Max	RMSE	Max	RMSE	Max	RMSE	Max	RMSE	Max	RMSE	Max	RMSE	Max
FAST-LIVO2	85.05	205.39	~~287.33~~	~~401.99~~	~~297.39~~	~~558.99~~	×	×	**0.22**	0.64	~~18.34~~	~~34.18~~	~~18.11~~	~~33.97~~
FAST-LIVO2 w/ reg.	75.08 ↓ 9.97	180.86 ↓ 24.53	~~290.79~~ ~~↑ 3.46~~	~~412.25~~ ~~↑ 10.26~~	~~294.02~~ ~~↓ 3.37~~	~~559.70~~ ~~↑ 0.71~~	×	×	0.23 ↑ 0.01	**0.59** ↓ 0.05	~~7.00~~ ~~↓ 11.34~~	~~11.22~~ ~~↓ 22.96~~	~~8.52~~ ~~↓ 9.59~~	~~18.04~~ ~~↓ 15.93~~
FAST-LIVO2 Closest-view	68.41	166.30	~~281.63~~	~~397.90~~	~~291.29~~	~~552.72~~	×	×	0.26	0.67	0.28	1.06	0.94	2.57
FAST-LIVO2 Closest-view w/ reg.	**67.47** ↓ 0.94	**163.69** ↓ 2.61	~~291.19~~ ~~↑ 9.56~~	~~412.67~~ ~~↑ 14.77~~	~~298.72~~ ~~↑ 7.43~~	~~562.81~~ ~~↑ 10.09~~	×	×	**0.22** ↓ 0.04	0.60 ↓ 0.07	0.25 ↓ 0.03	**1.01** ↓ 0.05	0.75 ↓ 0.19	1.91 ↓ 0.66
Ours w/o robust weighting	69.79	170.96	113.85	209.48	266.32	499.77	0.38	2.55	0.27	0.72	0.22	1.28	0.25	1.55
Ours w/o robust weighting w/ reg.	69.09 ↓ 0.70	166.41 ↓ 4.55	108.36 ↓ 5.49	198.84 ↓ 10.64	260.75 ↓ 5.57	**490.55** ↓ 9.22	**0.25** ↓ 0.13	**1.78** ↓ 0.77	0.26 ↓ 0.01	0.73 ↑ 0.01	0.21 ↓ 0.01	1.25 ↓ 0.03	**0.23** ↓ 0.02	1.50 ↓ 0.05
Ours w/o reg.	72.56	174.78	109.26	196.15	263.73	496.77	0.29	2.26	0.30	1.31	0.23	1.35	0.24	3.04
Ours	69.84 ↓ 2.72	172.44 ↓ 2.34	**99.57** ↓ 9.69	**179.35** ↓ 16.80	**260.41** ↓ 3.32	497.95 ↑ 1.18	0.30 ↑ 0.01	2.19 ↓ 0.07	0.25 ↓ 0.05	0.72 ↓ 0.59	**0.20** ↓ 0.03	1.20 ↓ 0.15	0.25 ↑ 0.01	**1.45** ↓ 1.59

~~Struck-out~~ entries correspond to runs whose odometry estimate is already inaccurate, so the effect of the regularization cannot be assessed meaningfully from them.

- Loss of wide-baseline coverage: our criterion selects the reference patch whose induced warp is closest to a square, which in practice favors a reference view close to the current one. The consequence is a narrower baseline between the reference and the current view, and therefore a weaker multi-view constraint on the reconstructed geometry, which can let drift accumulate over long trajectories. The normalized cross-correlation criterion of FAST-LIVO2 does the opposite, favoring the patch with the largest agreement across the views that observed the point, which yields wider view coverage. Where the affine model is trustworthy this is the better trade, and `Bridge01` and `Cave02` in Table I of the main paper are the two sequences on which it shows. Our selection therefore gives up multi-view leverage in exchange for a well-conditioned warp, and that exchange pays off only where the warp is genuinely at risk.
- Blindness to photometric degradation: the squareness score is computed from the plane-induced homography, that is, from geometry alone, and never from image content. It therefore cannot see motion blur, defocus or exposure change, and the selection offers no protection against them. Making the selection aware of photometric quality, for instance through the gradient energy of the reference patch, is left as future work.
- Divergence when the number of tracked features is too small: If the number of tracked features is very low, the system can easily diverge. While typical visual SLAM algorithms trigger relocalization when tracking is lost due to insufficient features, the current LIVO system does not consider this.
- Immediate divergence following odometry errors: Once the odometry drifts, visual mapping becomes incorrect, causing the state estimation to diverge immediately. Therefore, mechanisms such as reinitializing the existing map based on drift detection appear necessary.

We plan to address these issues in future work.

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